

Test 1: Statistical Stress Test

This will determine if the performance gap between the adaptive and baseline models remains statistically stable across a larger sample of network topologies.

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MONTE CARLO AVERAGES (50 runs)

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Execution Time: 1945.99 seconds

Metric | Baseline | Adaptive

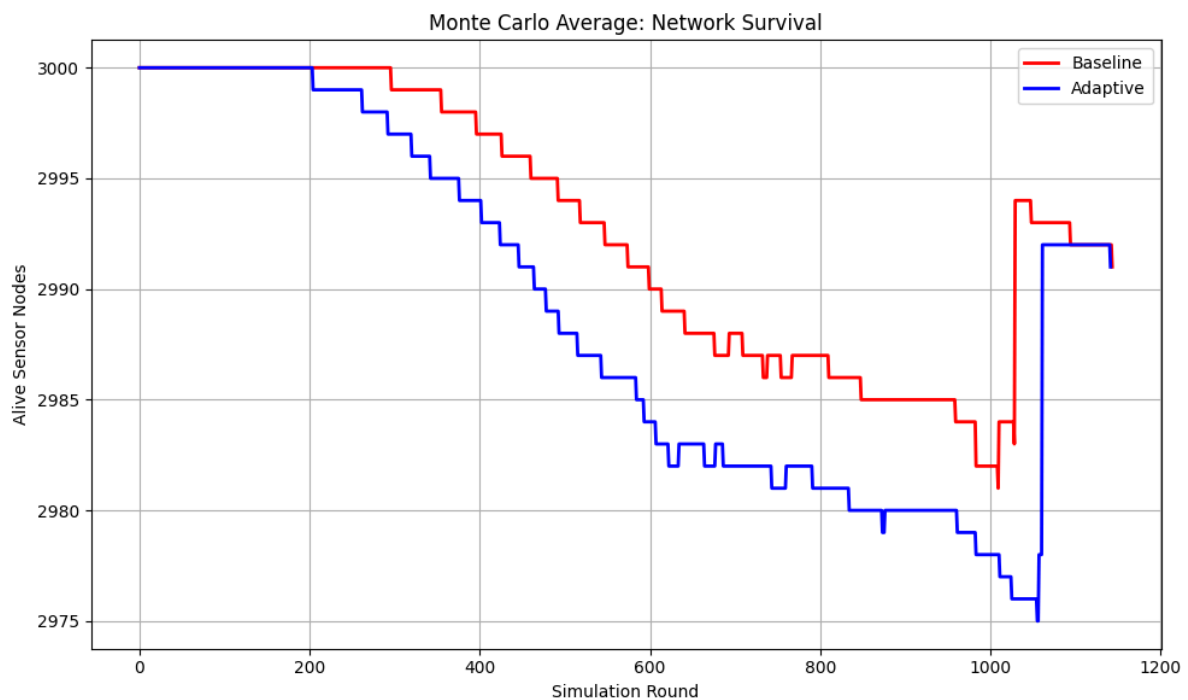
Avg Rounds Survived | 838.0 | 818.9

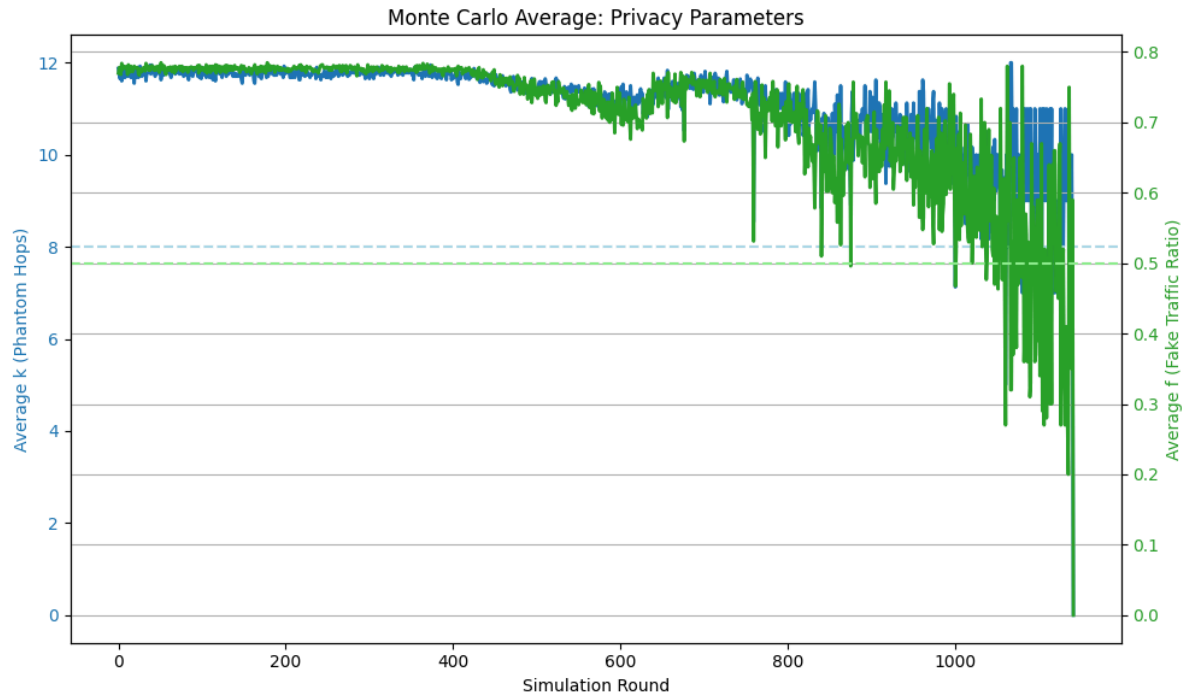
Avg Captures | 5.0 | 1.2

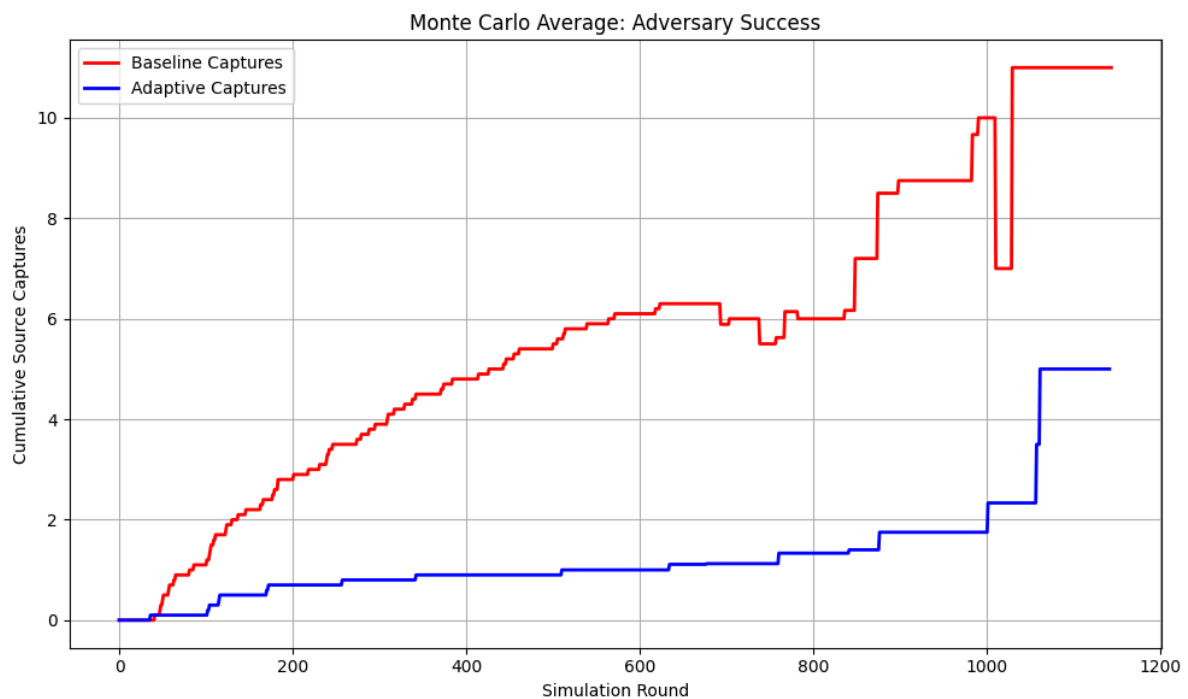
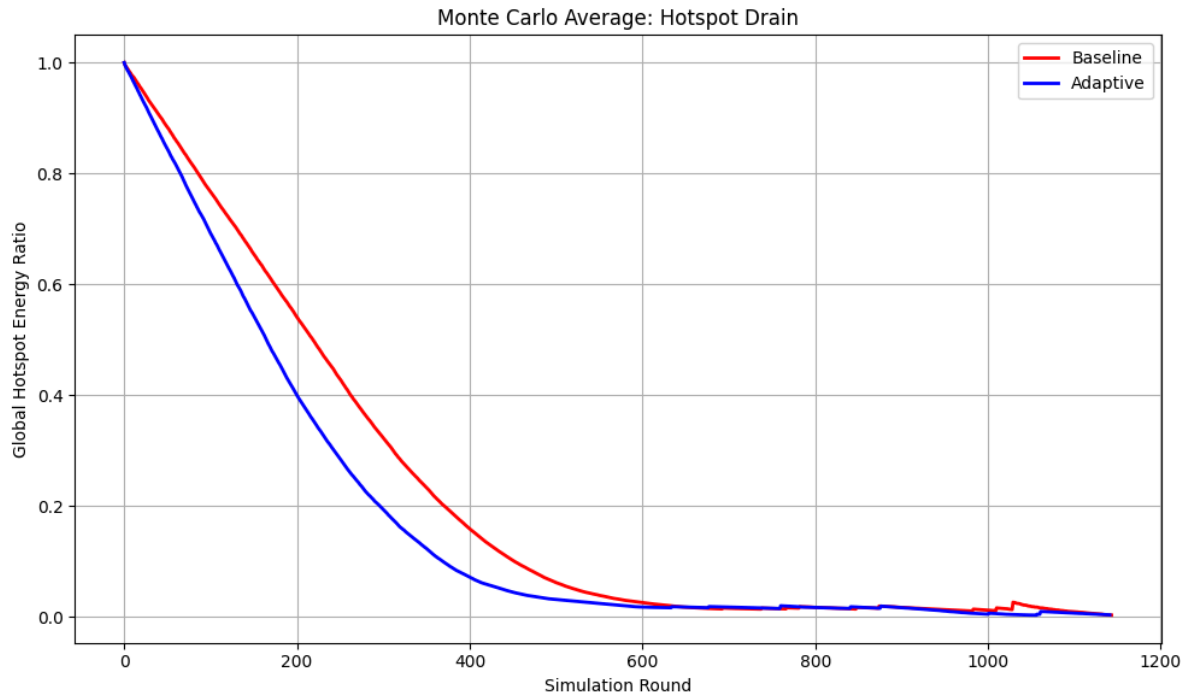
Avg Dead Nodes | 16.7 | 23.1

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Test 2: Density Stress Test.

To evaluate if the adaptive routing collapses when fewer physical pathways are available, we will test a sparse network topology.

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MONTE CARLO AVERAGES (10 runs)

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Execution Time: 106.77 seconds

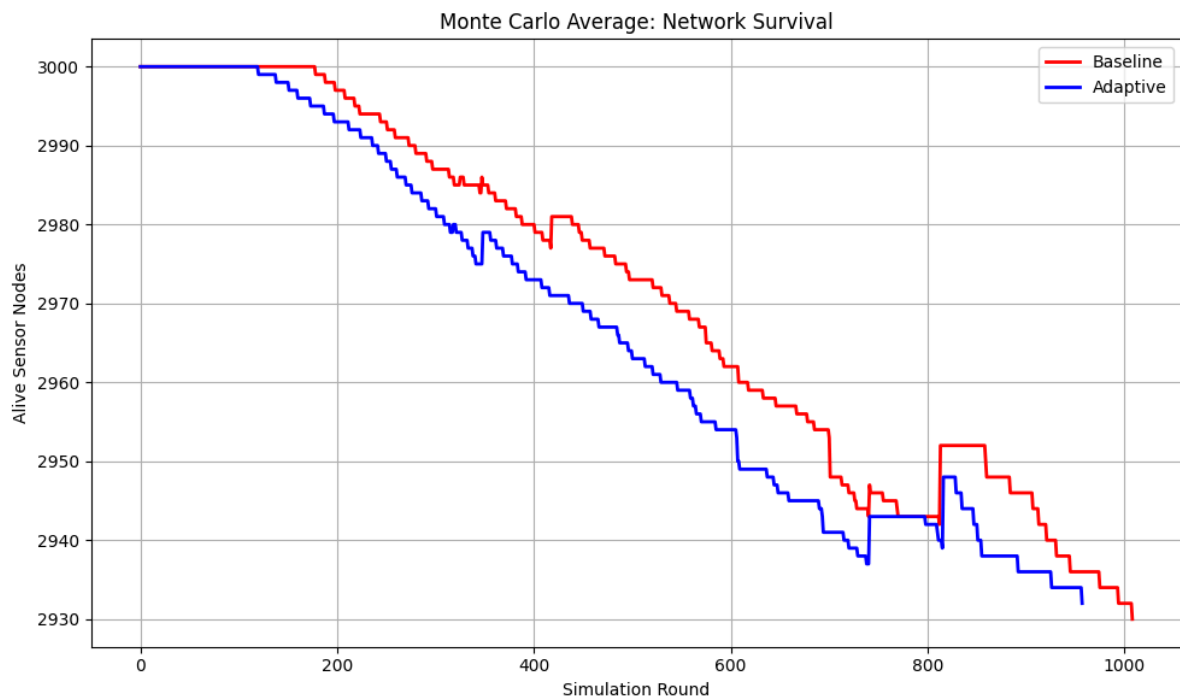
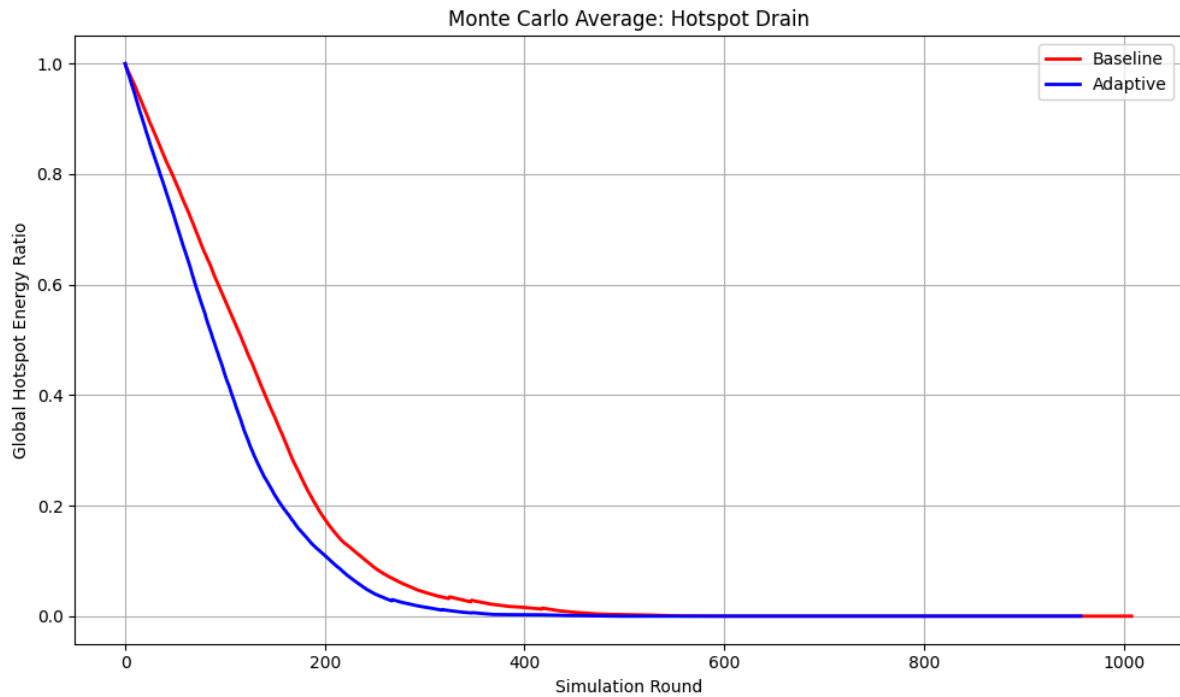
Metric	Baseline	Adaptive
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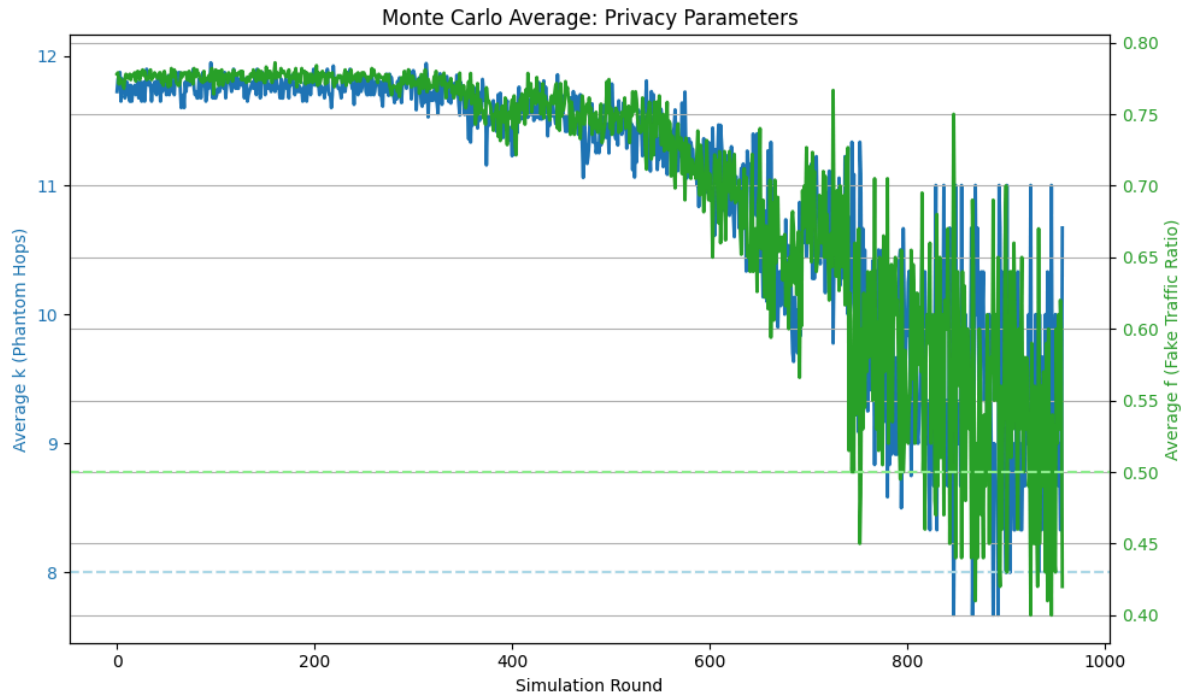
Avg Rounds Survived	624.7	598.9
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Avg Captures	2.8	1.2
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Avg Dead Nodes	21.9	25.0
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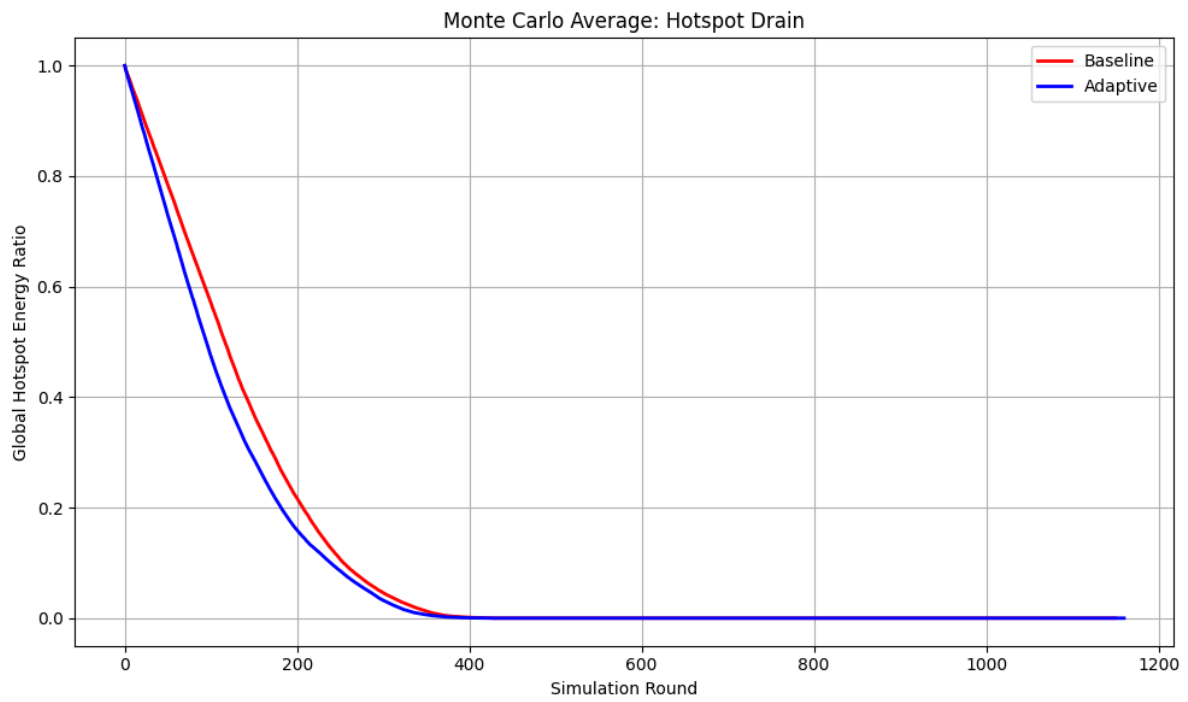
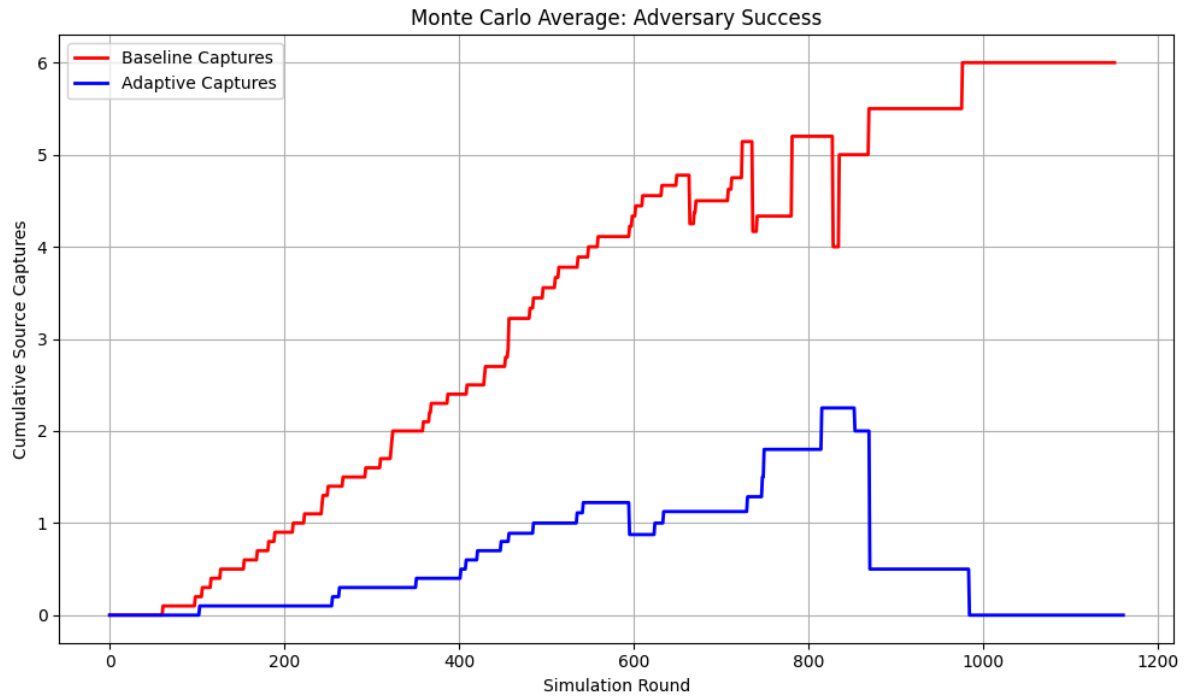


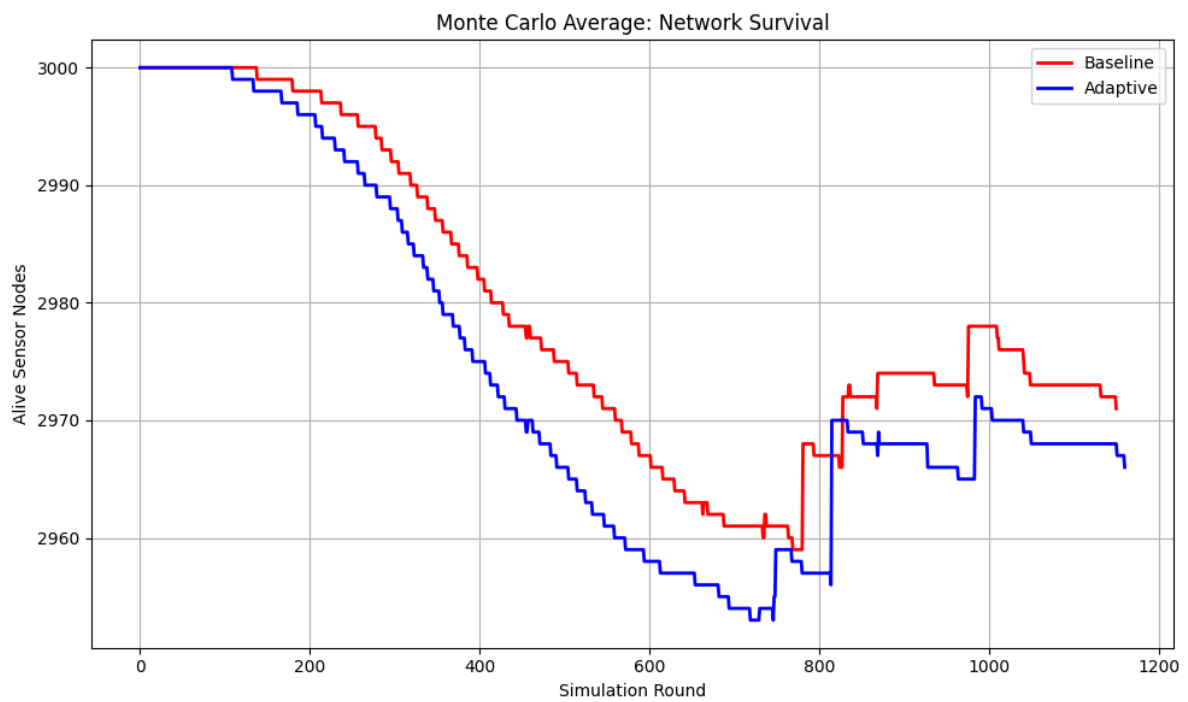
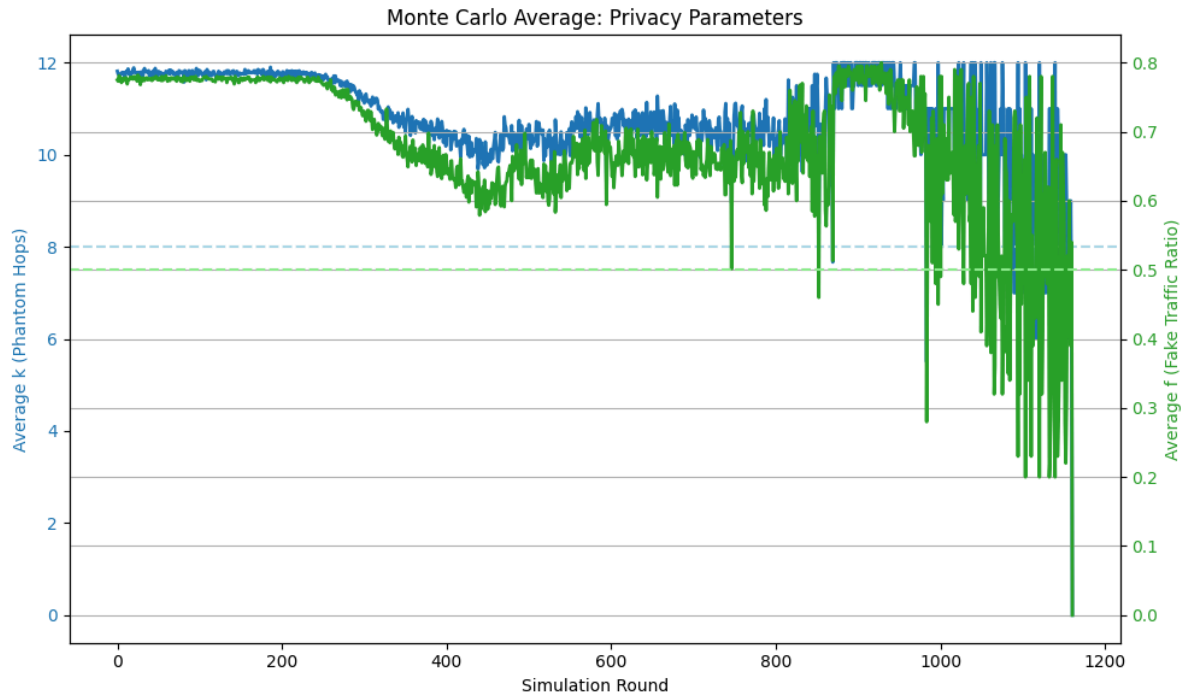


Test 3: Source Load Stress Test

We will increase the network load from 4 active sources to 8 to test the scalability of the Per-Source Optimization.

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MONTE CARLO AVERAGES (10 runs)
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Execution Time: 594.13 seconds
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Metric           | Baseline   | Adaptive
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Avg Rounds Survived | 801.1      | 795.1
Avg Captures       | 4.8        | 1.3
Avg Dead Nodes     | 43.8       | 50.5
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Test 4: Adversary Capability Stress Test.

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MONTE CARLO AVERAGES (10 runs)

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Execution Time: 594.13 seconds

Metric | Baseline | Adaptive

Avg Rounds Survived | 801.1 | 795.1

Avg Captures | 4.8 | 1.3

Avg Dead Nodes | 43.8 | 50.5

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